School of Natural Resources and the Environment
University of Arizona
Tucson, AZ, USA.

Fujiang Ji

Email: fujiang@arizona.edu | fujiang.ji@wisc.edu
Personal website: https://fujiangji.github.io/
Academic profiles: Google Scholar | ResearchGate | GitHub | ORCID

Research Interests

- ***Research Focus:*** How climate change and human disturbance reshape terrestrial ecosystem function?
- ***Research themes:*** Plant functional traits and diversity; forest fragmentation and edge effects; dryland biocrusts and climate feedback.
- ***Methods:*** Multiscale imaging spectroscopy; data fusion and machine learning; radiative transfer and process-based modeling.

Educational Background

- **Ph.D.** degree in **Forestry**, Department of Forest and Wildlife Ecology, University of Wisconsin-Madison, WI, USA, Sep. 2021— Dec. 2025.

  Research focused on transferable, cross-scale retrieval *of plant functional traits from leaf, airborne, and satellite spectroscopy. Best Ph.D. Dissertation Award in Forest Ecology (2026).*

- **M.S.** degree in ***Cartography and Geographic Information System***, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, Sep. 2018 – Jun. 2021.

  Research focused on *integrating process-based modeling, remote sensing, and data assimilation to estimate crop yield and available soil nutrients.*

- **B.Eng.** degree in ***Remote sensing science and technology***, Chengdu University of Technology, Chengdu, China, Sep. 2014 – Jun. 2018.

Academic Appointments

- **Postdoctoral Research Associate,** School of Natural Resources and the Environment, University of Arizona, Tucson, AZ, USA, January 2026 — Present.

- **Research Associate (unpaid appointment),** Department of Forest and Wildlife Ecology, University of Wisconsin-Madison, Madison, WI, USA, February 2026 — Present.
- **Research Assistant,** Department of Forest and Wildlife Ecology, University of Wisconsin-Madison, WI, USA, September 2021— December 2025.

Journal Articles

Peer-reviewed papers (* indicates corresponding author)

First / Corresponding Author

1. **Ji, F.;** Yang, J.; Townsend, P. A.; Zheng, T.; Kovach, K. R.; Yu, T.; Yang, R.; Liu, M.; Chen, M. Robust hyperspectral reconstruction from satellite and airborne observations via a deep hierarchical fusion network across heterogeneous scenarios, ***Remote Sensing of Environment***, 2026, 338, 115385. https://doi.org/10.1016/j.rse.2026.115385
2. **Ji, F.;** Zheng, T.; Shiklomanov, A. N.; Yang, R.; Townsend, P. A.; Li, F.; Hao, D.; Dashti, H.; Kovach, K. R.; You, H.; Zhou, J.; Chen, M. Tracking seasonal variability in plant traits from spaceborne PRISMA and NEON AOP across forest types and ecoregions, ***Remote Sensing of Environment***, 2026, 333, 115149. https://doi.org/10.1016/j.rse.2025.115149
3. **Ji, F.;** Li, F.; Dashti, H.; Hao, D.; Townsend, P. A.; Zheng, T.; You, H.; Chen, M. Leveraging transfer learning and leaf spectroscopy for leaf trait prediction with broad spatial, species, and temporal applicability, ***Remote Sensing of Environment***, 2025, 326, 114818. https://doi.org/10.1016/j.rse.2025.114818
4. **Ji, F.*;** Li, F.; Hao, D.; Shiklomanov, A. N.; Yang, X.; Townsend, P. A.; Dashti, H.; Nakaji, T., et al. Unveiling the transferability of PLSR models for leaf trait estimation: lessons from a comprehensive analysis with a novel global dataset, ***New Phytologist***, 2024, 243(1), 111–131. https://doi.org/10.1111/NPH.19807 (ESI Top 1% highly cited paper)
5. **Ji, F.;** Meng, J.; Cheng, Z.; Fang, H.; Wang, Y. Crop Yield Estimation at Field Scales by Assimilating Time Series of Sentinel-2 Data Into a Modified CASA-WOFOST Coupled Model, ***IEEE Transactions on Geoscience and Remote Sensing***, 2022, 60, 4400914 (pp. 1–14). https://doi.org/10.1109/TGRS.2020.3047102
6. **Ji, F.;** Meng, J.; Fang, H. Study on Soybean Yield Estimation Using the Coupled CASA and WOFOST Model, ***Remote Sensing Technology and Application***, 2020, 35(2), 406–415. (In Chinese with English Abstract). http://www.rsta.ac.cn/CN/Y2020/V35/I2/406

7. Wang, W.; Lu, B.; Li, Y.; **Ji, F.*** Multiscale Morphology-Based Detection of Shoreline Change Hotspots from Aerial Imagery Under Fluctuating Water Levels, ***Remote Sensing***, 2026, 18(8), 1148. https://doi.org/10.3390/rs18081148
8. Wang, W.; Lu, B.; Li, Y.; **Ji, F.*** A Quantitative Assessment of the Inconsistency Between Waterbody Segmentation and Shoreline Positioning in Deep Learning Models, ***Geomatics***, 2026, 6(1), 21. https://doi.org/10.3390/geomatics6010021

Co-author

9. You, H.; **Ji, F.;** Park, T.; Radeloff, V. C.; Hurtt, G.; Jiang, M.; Chen, M. Episodic changes in global forest edges in the 21st century, ***Nature Communications***, 2026, 17(1), 11352. https://doi.org/10.1038/s41467-026-78347-6
10. Smith, W. K.; Villarreal, M. L.; Lauria, C.; Rutherford, W. A.; Herrmann, S.; Scholl, V.; Howell, A.; Javadian, M.; **Ji, F.;** Zhang, F.; Burgess, M. A.; Kokaly, R.; Poulter, B.; Reed, S. C. Experimental evidence of a biological soil crust degradation climate warming amplification feedback, ***Communications Earth & Environment***, 2026, 7, 775. https://doi.org/10.1038/s43247-026-03874-5
11. Liu, H.; Xiao, J.; Hao, D.; Li, F.; **Ji, F.;** Chen, M. Importance of viewing angle: Hotspot effect improves the ability of satellites to track terrestrial photosynthesis, ***Remote Sensing of Environment***, 2025, 317, 114492. https://doi.org/10.1016/j.rse.2024.114492
12. Li, F.; Zhu, Q.; Yuan, K.; **Ji, F.;** Paul, A.; Lee, P.; Radeloff, V. C.; Chen, M. Projecting Large Fires in the Western US With an Interpretable and Accurate Hybrid Machine Learning Method, ***Earth's Future***, 2024, 12(10), e2024EF004588. https://doi.org/10.1029/2024EF004588
13. Zeng, Y.; Hao, D.; Park, T.; Zhu, P.; Huete, A.; Myneni, R.; Knyazikhin, Y.; Qi, J.; Nemani, R. R.; Li, F.; Huang, J.; Gao, Y.; Li, B.; **Ji, F.;** Köhler, P.; Frankenberg, C.; Berry, J. A.; Chen, M. Structural complexity biases vegetation greenness measures, ***Nature Ecology & Evolution***, 2023, 7(11), 1790–1798. https://doi.org/10.1038/s41559-023-02187-6 [Link1, Link2, Link3, Link4, Link5, Free version]
14. Zhang, S.; Dai, X.; Li, J.; Gao, X.; Zhang, F.; Gong, F.; Lu, H.; Wang, M.; **Ji, F.;** Wang, Z.; Peng, P. Crop classification for UAV visible imagery using deep semantic segmentation methods, ***Geocarto International***, 2022, 37(25), 10033–10057. https://doi.org/10.1080/10106049.2022.2032387
15. Dai, X.; He, X.; Guo, S.; Liu, S.; **Ji, F.;** Ruan, H. Research on hyper-spectral remote sensing image classification by applying stacked de-noising auto-encoders neural network, ***Multimedia Tools and Applications***, 2021, 80(14), 21219–21239. https://doi.org/10.1007/s11042-021-10735-0
16. Cheng, Z.; Meng, J.; **Ji, F.;** Wang, Y.; Fang, H.; Yu, L. Aboveground biomass estimation of late-stage maize based on the WOFOST model and UAV observations, ***Journal of Remote Sensing***, 2020, 24(11), 1403–1418. (In Chinese with English Abstract). https://doi.org/10.11834/jrs.20200069

17. Mao, H.; Meng, J.; **Ji, F.;** Zhang, Q.; Fang, H. Comparison of Machine Learning Regression Algorithms for Cotton Leaf Area Index Retrieval Using Sentinel-2 Spectral Bands, ***Applied Sciences***, 2019, 9(7), 1459. https://doi.org/10.3390/app9071459

In review/preparation

1. **Ji, F.;** Zheng, T.; Townsend, P. A.; Kovach, K. R.; Liu, M.; Chen, M. Plant functional trait estimation from PlanetScope–EMIT reconstructed high-resolution hyperspectral imagery reveals spectral-spatial fidelity trade-offs, ***Remote Sensing of Environment***, 2026. (Under review).
2. **Ji, F.*;** Reed, S. C.; Villarreal, M. L.; Lauria, C.; Rutherford, W. A.; Chen, M.; Howell, A.; Smith, W. K. Hyperspectral reflectance reveals climate change–driven successional regression in dryland biocrust communities, ***Global Change Biology***, 2026. (Under review).
3. **Ji, F.*;** Reed, S. C.; Villarreal, M. L.; von Nonn, J.; Lauria, C.; Chen, M.; Yan, D.; Herrmann, S.; Karnieli, A.; Rodríguez-Caballero, E.; Weber, B.; Smith, W. K. New opportunities for global monitoring of vulnerable biological soil crust, 2026. (In preparation).
4. **Ji, F.*;** You, H.; Smith, W. K.; Townsend, P. A.; Xu, X.; Zheng, T.; Hao, D.; Li, F.; Liu, S.; Chen, M. Forest edges impose a global syndrome on canopy function, 2026. (In preparation).
5. You, H.; Park, T.; Che-Castaldo, C.; Hao, D.; Chatterjee, A.; Li, F.; Wang, Z.; **Ji, F.;** Sen, B.; Wang, Y.; Lazzara, M.; Liu, Y.; Ball, B.; Lauret, N.; Chavanon, E.; Chen, M. Satellite-observed greenness expansion across Antarctica, ***Science***, 2026. (Under review).
6. Zhou, J.; Zhang, C.; You, H.; **Ji, F.;** Yang, Q.; Fan, W.; Li, X.; Qiao, L.; Yang, C.; Mulla, D.; Kumar, V.; Jin, Z.; Liu, L. Leveraging cross-sensor LiDAR observations and Earth embeddings for canopy height prediction, ***Remote Sensing of Environment***, 2026. (Under review).
7. Karnieli, A.; Chen, J.; Herrmann, S.; **Ji, F.;** Marais, E.; Martinez-Sanchez, J. F.; Noy, K.; Panov, N.; Rodríguez-Caballero, E.; Silver, M.; Reed, S.; Smith, W. K.; Villarreal, M.; Weber, B.; Zaady, E. From biocrust organic constituents to spaceborne mapping: A transferable hyperspectral framework for drylands, ***ISPRS Journal of Photogrammetry and Remote Sensing***, 2026. (Under review).
8. Yan, D.; Ryan, J. X.; Reed, S. C.; Villarreal, M. L.; **Ji, F.;** Karnieli, A.; He, J.; Smith, W. K. Reflectance spectroscopy enables robust classification of biocrust successional stages under dry and wet conditions, ***Environmental Research: Ecology***, 2026. (Under review).

Invited Talks & Conference Presentations

1. **Ji, F.;** NASA ARID seminar talk: Understanding the Dynamics of Plant Functional Traits Across Spatial, Temporal, and Biological Scales Using Multi-Source Hyperspectral Remote Sensing Observations (**Invited**, Apr. 2026).
2. **Ji, F.;** NASA's EMIT science team seminar: Hyperspectral reconstruction from spaceborne and airborne observations to map plant functional traits in fine detail (**Invited,** Jul. 2025).
3. **Ji, F.;** et al. Robust hyperspectral reconstruction from satellite and airborne observations via a deep hierarchical fusion network across heterogeneous scenarios, *AGU Annual Meeting 2025* (New Orleans, LA Dec. 2025).
4. **Ji, F.;** et al. Assessing the suitability of reconstructed high-resolution hyperspectral imagery from PlanetScope and EMIT for plant functional trait estimation, *AGU Annual Meeting 2025* (New Orleans, LA Dec. 2025).
5. **Ji, F.;** Zheng, T.; Yang, R.; Kovach, K.R.; Townsend, P. A.; Dashti, H.; Chen, M.; Seasonal variations of plant traits from PRISMA hyperspectral imagery over multiple ecological functional areas, *AmericaView Annual Meeting* (Madison, U.S., April., 2025).
6. **Ji, F.;** Zheng, T.; Yang, R.; Kovach, K.R.; Townsend, P. A.; Dashti, H.; Chen, M.; Seasonal variations of plant traits from PRISMA hyperspectral imagery over multiple ecological functional areas, *AGU Annual Meeting 2024* (Washington, D. C., U.S., Dec. 2024).
7. **Ji, F.;** You, H.; Chen, M. Unveiling the transferability of PLSR models for leaf trait estimation: lessons from a comprehensive analysis with a novel global dataset, Bryson Scholarship Poster Session, University of Wisconsin-Madison (Madison, U.S., Feb. 2024).
8. **Ji, F.** Eyes in the sky: Decoding plant functional traits with imaging spectroscopy. Three Minute Thesis competition, University of Wisconsin-Madison (Madison, U.S., Sep. 2024).
9. **Ji, F.;** Li, F.; Hao, D.; Chen, M., et al. Estimating Leaf Functional Traits with Leaf Spectroscopy and Physics-guided Transfer Learning Based Physical Model Across Biomes, *AGU Fall Meeting 2022* (Chicago, U.S., Dec. 2022).
10. Chen, M.; **Ji, F.;** Hao, D.; Zeng, Y.; et al. Fast estimation of leaf biochemical properties by inverting a simple leaf spectra model, *AGU Fall Meeting 2022* (Chicago, U.S., Dec. 2022).
11. You, H.; **Ji, F.;** Chen, M. Global Mappings of 21st-century Forest Edge Dynamic. Bryson Scholarship Poster Session, University of Wisconsin-Madison (Madison, U.S., Feb. 2024).

Research Experiences and Projects

- ***Postdoctoral Research Associate, University of Arizona, Jan. 2026 – Present.***

  1. **NASA Earth Surface Mineral Dust Source Investigation (EMIT) – Dryland Ecosystem Mapping with Multiscale Hyperspectral Remote Sensing**, *Jan. 2026 – Present.*

     Integrating hyperspectral observations from NASA's EMIT platform with complementary drone-, aircraft- (GEMx), and satellite-scale (EnMAP, PACE) datasets to map dryland plant and soil community composition and functional diversity across the Colorado Plateau and global drylands, with emphasis on developing novel maps of biological soil crust cover, community composition, and function.

- ***Graduate Research Assistant, University of Wisconsin-Madison, Sep. 2021 – Dec. 2025.***

  2. **Advance spaceborne mapping of plant functional traits with high-resolution and hyperspectral data over sparse vegetation canopies (80NSSC24K0054),** *Mar. 2024 – Dec. 2025.*

     Developed a data fusion framework to enable large-scale, repeatable mapping of plant functional traits in sparsely vegetated areas by combining PlanetScope imagery with DESIS or EMIT imaging spectroscopy.

  3. **Monitoring and understanding seasonal variations of forest functional traits and diversity by integrating observations from multi-source RS data,** *Sep. 2021– Dec. 2024.*

     Combined PRISMA imaging spectroscopy, NEON AOP observations, field measurements of leaf spectra and traits, and empirical, physical, and hybrid models to investigate how plant functional traits vary through the growing season and across forest ecosystems.

- ***Graduate Research Assistant, Aerospace Information Research Institute, Chinese Academy of Sciences, Sep. 2018 – Jun. 2021.***

  1. **China High-resolution EO System – Quantitative Retrieval Technology of Vegetation Parameters from GF-6 WFV Satellite Image (30-Y20A03-9003-17/18-05)**, *2018-2019.*

     Used GF-6 WFV observations to estimate crop yield in the Xinjiang study area, integrating crop growth modeling and data assimilation in IDL.

  2. **The STS (Science and Technology Service Network Initiative) Program of Chinese Academy of Sciences (KFJ-EW-STS-069)**, *2019-2020.*

     Conducted field campaigns and remote sensing monitoring to assess crop physiology, biochemistry, condition, biomass, soil nutrients, and yield. Organized monthly project meetings and drafted progress reports.

  3. **"Big Earth Data" Science Engineering Project of Chinese Academy of Sciences (CASEarth) –**

Big Earth Data Supports the U.N. Sustainable Development Goals (SDGs), *2020.*

Produced the 2000-2019 farmland productivity dataset in Northeast Eurasia by using the crop growth model through the JavaScript API interface of the Google Earth Engine platform; wrote the SDG 2.4 documents in both Chinese and English.

4. **Precision Insurance of Wheat Based on Spatial Big Data**, *2018-2019.*

   Assimilated time-series remote sensing observations into crop growth models to estimate yield and assess crop damage, and developed a technical framework for wheat insurance.

- ***Undergraduate Research Assistant, Chengdu University of Technology, Sep. – Dec. 2016.***

1. Research on technologies used to demarcate red-line areas of ecology in major districts and counties of Sichuan Province, China, initiated by a professor in the department, 2016.

- ***Project Leader, Chengdu University of Technology, 2017 – 2018.***

1. **National College Students' innovation and entrepreneurship training program (Grant No. 201710616032),** *2017-2018.*

   Inversion and Detection of Parameter of the Growing Status of Rice based on Hyperspectral Data.

Teaching & Workshops

- ***Guest Lecturer*** *— Nov. 5, 2026*

  BSE 301: Land Information Management, University of Wisconsin–Madison

  Understanding Terrestrial Ecosystems Through Remote Sensing -- Insights from Structure and Function in Forests and Drylands: Delivered a guest lecture on remote sensing applications in terrestrial ecosystem monitoring

- ***Workshop Co-Instructor*** *— Feb. 2024*

  Getting Started with Python for Analyzing Large Climate and Satellite Data

  UW–Madison Research Bazaar, University of Wisconsin–Madison, Madison, WI, USA

  Co-taught a workshop introducing Python-based approaches for analyzing large climate and satellite datasets (Workshop materials).

Honors and Awards

Postdoctoral Research Development Grant (University of Arizona Sursum Fellows) [Link]	2026
Best Ph.D. dissertation Award of Forest Ecology, UW-Madison [Link]	2026
Annual Joon Lee Award at Dept of Forest & Wildlife Ecology, UW-Madison [Link]	2025
Annual Thomas O'Brien Award at Dept of Forest & Wildlife Ecology, UW-Madison [Link]	2024

Excellent Student of University of Chinese Academy of Sciences [Link]	2020
Postgraduate student Scholarship of University of Chinese Academy of Sciences	2020
Outstanding Graduate (Class of 2018), Chengdu University of Technology [Link]	2018
Excellent Student of Chengdu University of Technology	2017
First prize of the 6th National College Student GIS Application Skills Competition [Link]	2017
Excellent Graduates of Sichuan Province, China [Link]	2017
Excellent Student of Chengdu University of Technology	2016
National Encouragement Scholarship	2015

Technical Skills

- **Programming:** Python, R, ENVI/IDL, MATLAB, JavaScript, Linux shell, Git.
- **Models:** World Food Studies (WOFOST), AquaCrop, PROSPECT, PROSAIL, Leaf-SIP.
- **Computing:** High-throughput computing (HTC), high-performance computing (HPC).
- **GIS and Remote sensing software:** GEE, ArcGIS, QGIS, ENVI, SNAP, ERDAS.
- **Statistical Analysis:** Regression, Machine learning, Deep learning, Network analysis.
- Skilled in remote sensing data processing, RS algorithm design and system development.
- Field instrumentation: LAI-2200, SPAD-502, and TDR-300.

Software Development

- ***hyperproc* — Python Package for Hyperspectral Remote Sensing**
  Developed a Python package providing a unified interface for searching, reading, and processing airborne and satellite hyperspectral data to support ecosystem monitoring workflows (Documentation | GitHub)

Service

- **Reviewer for scientific Journals:** *Remote Sensing of Environment; Agricultural and Forest Meteorology; Earth System Science Data; Remote Sensing; IEEE Transactions on Geoscience and Remote Sensing*; *Science of the Total Environment, Frontiers of Earth Science*.
- American Geophysical Union (AGU) session convener and chair.
- NASA proposal review panelist.

Professional Affiliation

- Student Member of American Geophysical Union (AGU), 2022-2023. 2024-present.

References

The following referees can provide recommendation letters:

1. **Min Chen, Advisor**
   Title: Associate Professor
   Institution: Department of Forest & Wildlife Ecology, University of Wisconsin-Madison.
   Email: min.chen@wisc.edu

2. **William Smith, Postdoc Advisor**
   Title: Associate Professor
   Institution: School of Natural Resources and the Environment, University of Arizona
   Email: wksmith@arizona.edu

3. **Philip A. Townsend, Doctoral Committee Member**
   Title: Professor
   Institution: Department of Forest & Wildlife Ecology, University of Wisconsin-Madison.
   Email: ptownsend@wisc.edu

4. **Alexey N. Shiklomanov, Doctoral Committee Member**
   Title: Research Physical Scientist
   Institution: NASA Goddard Space Flight Center.
   Email: alexey.shiklomanov@nasa.gov